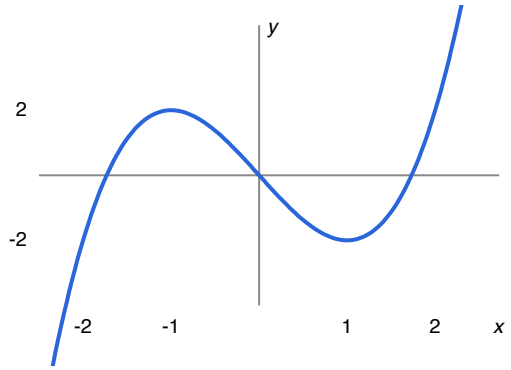
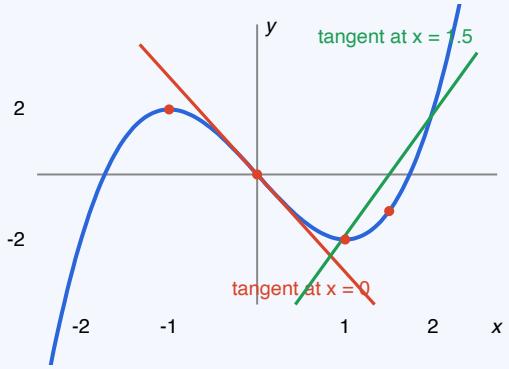
Graphs 8 — Gradient of a curve · Translating, reflecting and stretching graphs

Calculator allowed; graph paper for question 2. Marks in brackets. ★ = harder question. Key: blue = $y = \mathrm{f}(x)$, red = first transformed graph, green = second.

1 The graph of $y = x^3 - 3x$ is shown. **a** By drawing tangents, estimate the gradient of the curve at **i** $x = 0$ **ii** $x = 1.5$. **b** Where is the gradient zero? [6]



Draw a straight line touching the curve at the point, then gradient $= \dfrac{\text{rise}}{\text{run}}$ from the tangent.



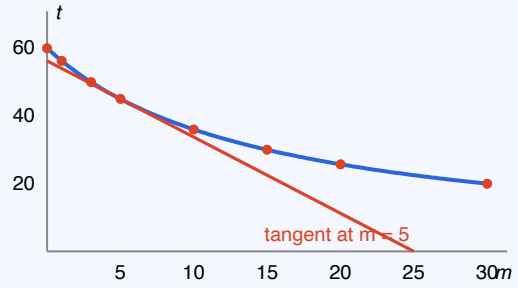
a i At $x = 0$ the tangent slopes down: gradient ≈ -3. **ii** At $x = 1.5$ it slopes steeply up: gradient ≈ 3.8 (anything from about 3.5 to 4 is a reasonable estimate). **b** At the turning points, $x = -1$ (maximum) and $x = 1$ (minimum).

2 The temperature t °C of a cup of tea, m minutes after it is poured, is modelled by $t = \dfrac{900}{m + 15}$ for $0 \le m \le 30$. **a** Copy and complete the table (nearest degree). **b** Draw the graph of t against m. **c** Find the rate at which the tea is cooling, in °C per minute, when $m = 5$. [5]

m (min)	0	1	3	5	10	15	20	30
t (°C)	60			45		30		20

a

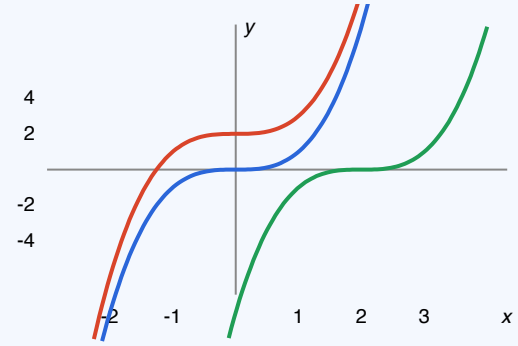
m (min)	0	1	3	5	10	15	20	30
t (°C)	60	56	50	45	36	30	26	20



c Draw the tangent at $m = 5$ and find its gradient: about -2.2 to -2.3, so the tea is cooling at roughly 2.2 °C per minute at that moment (negative gradient = temperature falling).

3 Here is a sketch of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x^3$. **a** Sketch **i** $y = \mathrm{f}(x) + 2$ **ii** $y = \mathrm{f}(x - 2)$. **b** Write down where the point $(0, 0)$ is mapped to on each graph. [4]

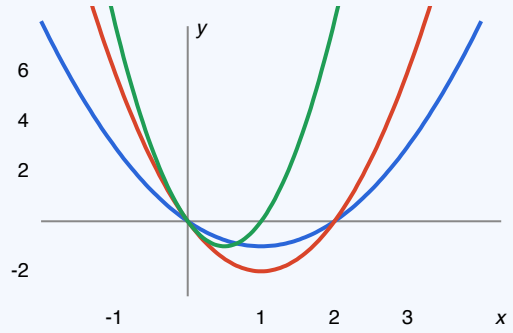
a $\mathrm{f}(x) + 2$ translates the graph *up* by 2 (red); $\mathrm{f}(x - 2)$ translates it *right* by 2 (green) — the sign inside the bracket works the opposite way to what it looks like.



b $(0, 0) \to (0, 2)$ for the first graph and $(0, 0) \to (2, 0)$ for the second.

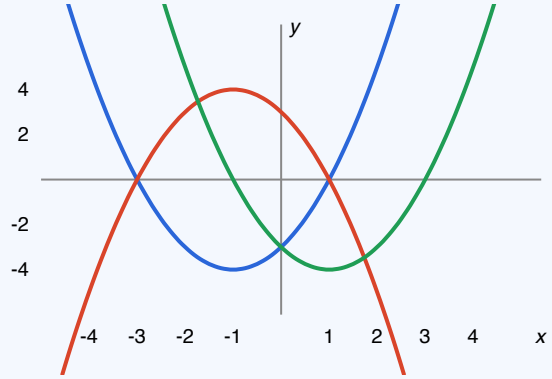
4 The graph of $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = (x-1)^2 - 1$, has vertex $(1, -1)$ and crosses the x-axis at 0 and 2. Sketch **a** $y = 2\mathrm{f}(x)$ **b** $y = \mathrm{f}(2x)$, marking the vertex and intercepts of each. [4]

a $2\mathrm{f}(x)$: every y-coordinate is doubled — a stretch parallel to the y-axis, scale factor 2. Vertex $(1, -2)$; x-intercepts unchanged at 0 and 2 (red).
b $\mathrm{f}(2x)$: every x-coordinate is halved — a stretch parallel to the x-axis, scale factor $\frac{1}{2}$. Vertex $(0.5, -1)$; intercepts 0 and 1 (green).



5 The diagram shows $y = \mathrm{f}(x)$ with $\mathrm{f}(x) = (x+1)^2 - 4$ (vertex $(-1, -4)$, x-intercepts -3 and 1). **a** On the same axes sketch **i** $y = -\mathrm{f}(x)$ **ii** $y = \mathrm{f}(-x)$. **b** Describe the transformation from $y = \mathrm{f}(x)$ to $y = -\mathrm{f}(x)$. **c** Describe the transformation from $y = \mathrm{f}(x)$ to $y = \mathrm{f}(-x)$. [6]

a i $-\mathrm{f}(x)$ flips every y-value: an upside-down parabola with vertex $(-1, 4)$ and the same intercepts -3, 1 (red). **ii** $\mathrm{f}(-x)$ flips every x-value: vertex $(1, -4)$, intercepts -1 and 3 (green).



b A reflection in the x-axis. **c** A reflection in the y-axis.

Total: 25 marks

Check: gradient at a point = gradient of the tangent (rise ÷ run, with sign) · $\mathrm{f}(x) + a$ up, $\mathrm{f}(x + a)$ LEFT, $\mathrm{f}(x - a)$ right · $-\mathrm{f}(x)$ reflect in x-axis, $\mathrm{f}(-x)$ reflect in y-axis · $k\mathrm{f}(x)$ stretch y by k; $\mathrm{f}(kx)$ stretch x by $\frac{1}{k}$ · always say which axis and the scale factor or vector.